Hadrien Tramoni

Quantitative Developer – C++, derivatives pricing, Monte Carlo on GPU

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github.com/HadrienT tramonihadrien.com French citizen, EU work rights, open to relocation

QUANTITATIVE DEVELOPMENT

quant-modeling – C++ Derivatives Pricing Library github.com/HadrienT/quant-modeling

Independent full-time work – live demo at quant.tramonihadrien.com *Nov. 2024 – Present*

- Studied stochastic calculus, volatility modelling and numerical methods full time (Shreve, Gatheral, Bergomi, Glasserman, Savine), then built the library below from January 2026.
- **Pricing library in C++20** that keeps instruments, models and engines separate: closed-form, binomial and trinomial trees, Crank-Nicolson PDE and Monte Carlo, for about 20 products across equity, rates, FX and commodities.
- **Adjoint algorithmic differentiation (AAD)** on an operator-overloading tape: the full vector of Greeks for roughly the cost of one extra valuation, instead of one bump per input.
- **Monte Carlo on GPU (CUDA)**, two Tesla V100, bit-identical across GPU counts and within 10^{-12} of the CPU: a daily-monitored barrier option under local volatility reaches a 10^{-4} relative standard error in 12.7 s, against 27 min on 8 CPU threads.
- **Volatility calibration** on stored option chains: SVI per maturity, Dupire local volatility, Heston (COS pricing, Levenberg-Marquardt) and stochastic-local volatility by the particle method.
- **Payoff scripting language** (lexer, parser, AST), so that one generic Monte Carlo engine prices any scripted payoff, and an exposure engine computing the **CVA** of a collateralised netting set.
- **Tested by properties** in about 70 GoogleTest suites (put-call parity, in + out = vanilla, measured convergence order) and under ASan/UBSan; exposed through pybind11, FastAPI and a React front end, self-hosted with Docker and GitHub Actions CI.
- Built alongside the study above, with an AI coding assistant (Claude Code).

PROFESSIONAL EXPERIENCE

Richemont International SA Geneva, Switzerland

Machine Learning Engineer *Nov. 2023 – Nov. 2024*

- Built a computer vision system that reads serial numbers from product images: designed the experiments, iterated on the models and reached over 90% accuracy on the group's own products.
- Applied LLMs (PaLM, Gemini) to extract structured data from unstructured documents; fine-tuned and evaluated the models on Vertex AI.
- Took the prototypes to production as an API on GCP Cloud Run, with the infrastructure managed in Terraform.
- Acted as the link between two teams, supervised an intern and delegated work to a research assistant from the Swiss Data Science Center; presented the results at Watches and Wonders 2024.

EDUCATION

ENSSAT – École Nationale Supérieure des Sciences Appliquées et de Technologie Lannion, France

Engineering Degree (Diplôme d'Ingénieur, MSc level) in Computer Science *2020 – 2024*

IT University of Copenhagen Copenhagen, Denmark

Exchange semester – Machine Learning, Computer Science *Aug. 2022 – Jan. 2023*

Lycée Jean Dautet – Classes Préparatoires aux Grandes Écoles (CPGE) La Rochelle, France

Intensive undergraduate programme in Mathematics and Physics *2016 – 2020*

SKILLS

Programming:	C++ (C++20), Python, CUDA, C, SQL, TypeScript
Quantitative finance:	Monte Carlo and variance reduction (antithetic and control variates, Sobol, Brownian bridge), finite-difference PDE, trees, local and stochastic volatility, short-rate models (Vasicek, Hull-White), calibration, AAD, CVA
Tools:	CMake, GoogleTest, pybind11, Eigen, NumPy, pandas, PyTorch, FastAPI, PostgreSQL, Linux, Git, Docker, GitHub Actions, GCP, Terraform
Languages:	French (native), English (C1), German (B2)